THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The difference between the 20th and the 19th square numbers is: 1

(a) 39 **(b)** 20 **(c)** 41 **(d)** 37

2. The smaller angle between the hands of a clock at 1 o'clock is 30°. The smaller angle between the hands at 11 o'clock is: 1

(a) 330 **(b)** 60 **(c)** 30 **(d)** 150

3. The sum of 10 consecutive counting numbers starting from 12 is: 1

(a) 120 **(b)** 165 **(c)** 175 **(d)** 220

4. A school bus carries 30 children on trip 1, 45 on trip 2 and 60 on trip 3. A pictograph uses one symbol for 15 children. The number of symbols needed for all three trips is: 1

(a) 11 **(b)** 9 **(c)** 10 **(d)** 8

5. The greatest number that divides 48, 72 and 120 leaving no remainder is: 1

(a) 25 **(b)** 24 **(c)** 23 **(d)** 26

6. A pattern adds the same number every time. Its first term is 7 and its 14th term is 150. The number added each time is: 1

(a) 11 **(b)** 12 **(c)** 13 **(d)** 10

7. The minute hand of a clock takes how many minutes to turn through 150°? 1

(a) 5 **(b)** 30 **(c)** 12 **(d)** 25

8. A 5-digit number ends with digit 7 and has digit sum 31; the sum of its other four digits is: 1

(a) 33 **(b)** 24 **(c)** 15 **(d)** 38

9. Out of 120 students, 45 play cricket and 38 play football, none plays both. How many play neither? 1

(a) 37 **(b)** 47 **(c)** 75 **(d)** 82

10. For the 3-digit number 1d3, the digit d that makes it divisible by 9 (smallest such d) is: 1

(a) 14 **(b)** 5 **(c)** 7 **(d)** 6

11. A pattern adds 1, then 2, then 3, ... Starting at 10, the term after 12 additions is: 1

(a) 88 **(b)** 22 **(c)** 76 **(d)** 154

12. 4 lines are drawn through a point O. The number of straight angles at O (pairs of opposite rays) is: 1

(a) 3 **(b)** 4 **(c)** 12 **(d)** 5

13. **Assertion (A):** In a pictograph where one symbol stands for 10 children, 3 and a half symbols show 35 children. 1
Reason (R): A half symbol stands for half of the value of a whole symbol.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** The number 1 is neither prime nor composite. 1
Reason (R): The number 1 has only one factor, namely itself.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The sum of the first 5 odd numbers is 25. 1
Reason (R): 25 is a square number.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. The sum of the first n odd numbers is 961. Find n and the 31st odd number. 2

17. A ray rotates 2 full turns and 90° more. Find the total angle turned and the number of right angles in it. 2

18. Find the number of terms in the hailstone (Collatz) sequence of 11 and its largest term. 2

19. A bar graph uses 1 cm = 20 units and a bar is 9 cm. Another bar is 3 cm shorter. Find the value of the shorter bar. 2

20. Show that the product of 3 consecutive numbers starting from 14 is divisible by 3. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A row of 8 numbers starts 2, 3 and each next number is the sum of the previous two. Find the 8th number and the sum of all 8 numbers. 3

22. In a pictograph one symbol = 12. A shop sold 3 symbols of pens, and the number of notebooks is 2 times the pens. Find the notebooks sold and total items sold. 3

23. Find the greatest number that divides 76 and 100 leaving remainder 4 in each case. 3

24. A tower is built in rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 6, and how many in the first 4 rows together? 3

OR

A pizza is cut by 2 lines through the centre into equal slices (all lines passing through the centre). Find the angle of each slice and the number of slices. 3

25. How many 4-digit numbers can be formed from 2, 5, 7, 9 using each digit once? What is the largest and the smallest? What is their difference? 3

OR

For the data 5, 5, 2, 7, 8, 3, 5, 1, 7, 4, 6, 5 prepare a frequency table and find how many values are greater than 5. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** The clock on the wall of Anil's school shows 2 o'clock. The hour hand turns slowly, and the minute hand moves faster. 5

(i) Through how many degrees does the minute hand turn in 20 minutes?

(ii) Through how many degrees does the hour hand turn in 1 hour?

(iii) What is the type of the angle between the hands at 4 o'clock?

(iv) Find the smaller angle between the hands at 7 o'clock and name the type of the other angle formed.

27. A robot starts facing North. It turns clockwise 180°, then 45° anticlockwise, then 90° clockwise. Find its final direction angle clockwise from North and the further clockwise turn needed to face North again. 5

OR

Find the number of steps the reverse-and-add process takes from 67 to reach a palindrome, write each sum, and state whether the final palindrome has an even or odd number of digits. 5

END OF QUESTION PAPER — Total: 50 Marks

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